Structuring with issue trees

How do we break a complex problem into questions we can actually answer?



Today

By the end of today, you will have

Built a three-level MECE issue tree for your case, written and sealed a working answer, and started your Analyst's Ledger.

- Apply the two MECE tests to a set of branches.
- Build an issue tree to three levels that ends in answerable questions.
- Distinguish a question-led tree from a hypothesis-led tree.
- Write a sealed working answer of no more than 100 words.

Competencies: B1 · B2 · B3 · F4

Session plan

10 min	**Opening problem**
20 min	**Concept: MECE and trees**
20 min	**Demonstration: Madhyapur tree**
35 min	**Activity: issue tree challenge**
25 min	**Own case: tree and sealed answer**
10 min	**Debrief and exit task**

Opening problem

- **Why do households in Madhyapur hand over mixed waste? List every reason you can think of in two minutes.**

Then look at your list. Do any reasons overlap? Is anything obviously missing?

MECE: two tests for every split

ME

Mutually exclusive

No item belongs in two branches. If awareness and attitude overlap, an argument about one is an argument about both, and the analysis double-counts.

CE

Collectively exhaustive

Together, the branches cover every possible answer. If a cause could exist outside your branches, the tree has a hole, and you may never look there.

MECE is associated with Barbara Minto's work at McKinsey. It disciplines the structure of questions; it does not claim that causes are independent.

A question-led issue tree for Madhyapur



The split follows the waste chain: home, collection, processing, enforcement. Every leaf is a question evidence could answer.

Four ways to split a problem

1

By stage

Follow the process: generation, collection, processing. Good for delivery problems.

2

By actor

Households, crews, contractor, corporation. Good when incentives differ.

3

Supply and demand

Is the service not offered, or not taken up? Good for low uptake.

4

By formula

Outcome equals coverage times uptake times quality. A driver tree, for outcomes that break down arithmetically.

Choose the split that best serves the decision. Two splits can both pass the MECE test and still differ in usefulness.

Question-led and hypothesis-led trees

Question-led

- Starts from the problem as a question
- Lists every branch that could matter
- Safer when you know little
- Risk: analysing everything

Hypothesis-led

- Starts from your best current answer
- Lists what must be true for it to hold
- Faster when you know something
- Risk: confirming what you already believe

Build the question-led tree first. Then write your best answer, and seal it.

The sealed working answer

- **Write your best current answer to your problem in no more than 100 words. Seal it. You will open it in your final reflection.**

Madhyapur, Day 3: households stop separating because crews mix the streams, and crews mix them because the contractor is paid by the tonne. Fixing the contract matters more than awareness. It may be wrong. That is the point.

The Analyst's Ledger starts today

Tag	Meaning	Madhyapur example
Fact	Stated in a source you can cite	The 2016 rules require segregation at source
Inference	Follows from facts by reasoning you can explain	Mixed loads suggest separation is lost at some stage
Assumption	Taken as true without evidence, for now	Households own enough containers
Hypothesis	A claim you intend to test	Payment by the tonne discourages separate collection

Tag every substantive claim. The Ledger is sampled at every gate (competency F4).

Activity: The issue tree challenge

Build a MECE tree under time pressure, then defend its logic against another pod.

Scenario

Suppose a state scheme that gives bicycles to girls entering secondary school is reported to have low uptake in some districts. The state department asks why. Build the tree before you know anything else.

What to do

1. Agree the question at the root in five minutes.
2. Choose a splitting logic and build level one. Test it for MECE.
3. Build two more levels until every leaf is an answerable question.
4. Swap trees with another pod. Find one overlap and one gap.
5. Repair your tree and note what changed.

Time 35 minutes

Groups Pods of four

Output A three-level tree with its splitting logic stated

Your case: Worksheet 2.1

Issue tree and sealed working answer

1. Write your problem as a question at the root of the tree.
2. Choose a splitting logic for level one and test it for MECE.
3. Build down to three levels, ending in answerable questions.
4. Write your working answer in no more than 100 words and seal it.
5. Start your Ledger: tag every claim on today's worksheet.

Hand in

Worksheet 2.1 by the start of Day 4. Send the sealed answer to the instructor separately; it is not marked and not opened until Day 20.

Quality check

Each level uses one splitting logic.

Tag every claim in your Analyst's Ledger: fact, inference, assumption or hypothesis.

What to take away

1. A tree is only as good as its splitting logic. Name it.
2. MECE has two tests: no overlaps and no gaps.
3. Leaves must be questions that evidence can answer, not solutions.
4. Write your best answer early, then try to break it.

Exit task Name your level-one splitting logic in five words.

Before the next session

Independent work

Show your tree to someone outside the course and ask them to find a gap. Revise it. Add one more answerable question to your weakest branch. About 40 minutes.

Read

Conn and McLean, Bulletproof Problem Solving (2018), the chapter on disaggregating problems with logic trees.

Next

Day 4: Finding causes

Tomorrow we move from questions to causes, using problem trees, Five Whys and the fishbone.